

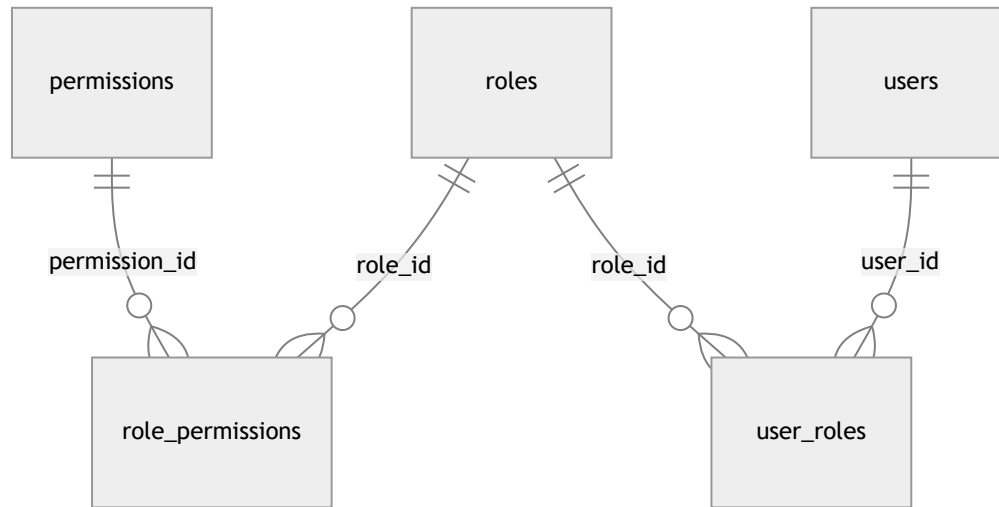
The 68 tables that hold data

Every table the seeded school actually uses, what each one is for, the columns that carry meaning, and an entity diagram per area. The full reference covers all 93 tables and all 1103 columns; this is the part you keep open.

Four columns are left out below because they repeat on almost every table: `id` , `school_id` , `created_at` and `updated_at` . Stating them once rather than 68 times removes 265 of 763 rows. `school_id` is the tenant key, so its edge is left off the diagrams too, which would otherwise draw it on nearly every table.

Four tables are exceptions, deliberately. `permissions` , `scheduled_jobs` , `schools` carry no `school_id` : they derive from `TimestampedBase` rather than `TenantBase` , because `schools` is the tenant, and the permission vocabulary and the job schedule belong to the software rather than to any one school. `alembic_version` has none of the four — it is Alembic's own bookkeeping and not part of the product.

Tenancy, identity and access



schools 1 rows

One customer. The only table without a `school\_id` of its own.

used by academic_years, admission_cycles, admission_decisions, admission_offers, application_guardians, application_medical and 83 more

COLUMN	TYPE	NOTES
code	varchar(16)	unique
name	varchar(160)	
status	varchar(10)	
address	text	optional
city	varchar(80)	optional
state	varchar(80)	optional
pincode	varchar(10)	optional
phone	varchar(20)	optional
email	varchar(160)	optional
website	varchar(200)	optional
logo_url	text	optional
primary_color	varchar(9)	optional
board	varchar(20)	optional
affiliation_no	varchar(40)	optional

academic_years 1 rows

A session. Several may be open at once — 2026-27 active while 2027-28 takes admissions and 2025-26 is closing is the normal state of a school in January, which the v0 string could not express.

used by admission_cycles, assessment_schemes, audit_log, class_sections, enrolments, fee_invoices and 4 more

COLUMN	TYPE	NOTES
code	varchar(9)	unique
start_date	date	
end_date	date	
status	varchar(15)	
is_current	boolean	
promotion_completed_at	date	optional
result_published_at	date	optional
closed_at	date	optional

users 221 rows

Every person who can sign in - staff, students and guardians alike. One row per login; the role-specific detail lives in `students`, `guardians` or `employees`, each of which points back here.

used by admission_decisions, application_payments, applications, assessments, audit_log, candidate_offers and 29 more

COLUMN	TYPE	NOTES
role	varchar(7)	
login_id	varchar(64)	unique
password_hash	varchar(255)	
full_name	varchar(120)	
email	varchar(160)	optional
phone	varchar(20)	optional
photo_url	text	optional
is_active	boolean	

roles 14 rows

A named set of permissions. System roles ship with the product and cannot be deleted; a school may copy one and adjust it.

used by role_permissions, user_roles

COLUMN	TYPE	NOTES
code	varchar(40)	unique
name	varchar(80)	
description	text	optional
is_system	boolean	

permissions 91 rows

A thing that can be done, named `module.resource.action`.

used by role_permissions

COLUMN	TYPE	NOTES
code	varchar(80)	unique
module	varchar(40)	
action	varchar(40)	
description	text	optional

role_permissions 325 rows

Which permissions a role grants. Per school, so two tenants can define the same role differently.

needs permissions, roles

COLUMN	TYPE	NOTES
role_id	bigint	→ roles.id, unique
permission_id	bigint	→ permissions.id, unique

user_roles 228 rows

A user holds a role, optionally narrowed to part of the school.

needs roles, users

COLUMN	TYPE	NOTES
user_id	bigint	→ users.id, unique
role_id	bigint	→ roles.id, unique
scope_type	varchar(13)	unique
scope_id	bigint	unique, optional

settings 3 rows

Per-school configuration as key/value pairs, with the vocabulary defined in `core/settings\_registry.py`. Module switches live here too, as `feature`.

COLUMN	TYPE	NOTES
key	varchar(80)	unique
value	json	

custom_fields 2 rows

One school-defined attribute on students, guardians, employees or applications (§3.15 level 2).

COLUMN	TYPE	NOTES
entity	varchar(11)	unique
key	varchar(40)	unique
label	varchar(120)	
field_type	varchar(7)	
options	json	optional
is_required	boolean	
is_active	boolean	
sort_order	integer	

number_sequences 3 rows

A gapless counter per school, per type, per year.

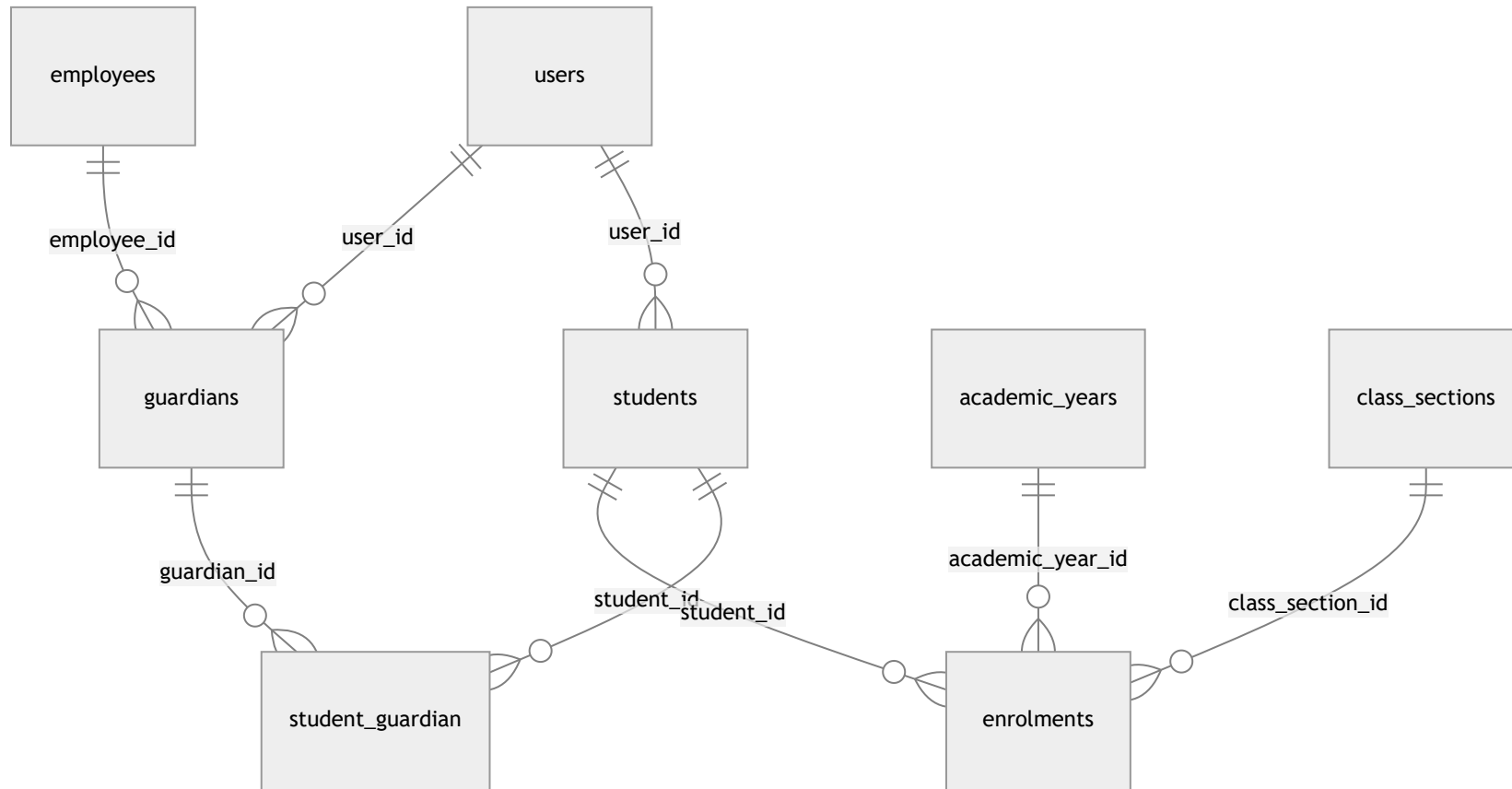
COLUMN	TYPE	NOTES
kind	varchar(24)	unique
year	integer	unique
prefix	varchar(24)	optional
width	integer	
next_value	integer	

alembic_version 1 rows

Alembic's own bookkeeping - the single row naming the migration this database is at. Not part of the product; never edit it by hand.

COLUMN	TYPE	NOTES
version_num	varchar(32)	

People - students and guardians



students 100 rows

What is true about a student for life. Which class they sit in is a fact about a *year* and lives on `enrolments` (ERP_BLUEPRINT §3.2).

needs users

used by application_siblings, applications, enrolments, grievances, message_recipients, student_guardian

COLUMN	TYPE	NOTES
user_id	bigint	→ users.id , unique
admission_no	varchar(32)	unique
status	varchar(15)	
dob	date	optional
gender	varchar(6)	optional
address	text	optional
admission_date	date	optional
custom	json	

guardians 98 rows

Whoever is responsible for a child — not necessarily a parent (§3.4).

needs employees, users **used by** message_recipients, student_guardian

COLUMN	TYPE	NOTES
user_id	bigint	→ users.id , unique
occupation	varchar(80)	optional
employee_id	bigint	→ employees.id , optional

student_guardian 100 rows

The link, with the structure §3.4 asks for: which relation, and which one of them the school actually rings.

needs guardians, students

COLUMN	TYPE	NOTES
guardian_id	bigint	→ guardians.id , unique
student_id	bigint	→ students.id , unique
relation	varchar(14)	
is_primary	boolean	

enrolments 100 rows

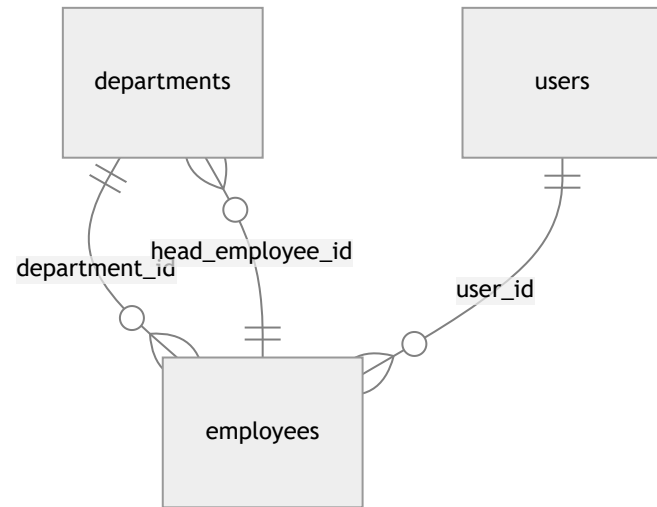
A student's membership of a section for one academic year. ****The most important join in the schema****: every year-scoped fact - fees, attendance, marks, a bus seat - hangs off `enrolment\_id`, never `student\_id`.

needs academic_years, class_sections, students

used by attendance, fee_concessions, fee_invoices, fee_payments, grievances, homework_submissions and 5 more

COLUMN	TYPE	NOTES
student_id	bigint	→ students.id
academic_year_id	bigint	→ academic_years.id
class_section_id	bigint	→ class_sections.id , unique
roll_no	integer	unique
status	varchar(15)	
joined_on	date	optional
left_on	date	optional
house	varchar(20)	optional

People - staff



employees 16 rows

Any member of staff, teaching or not (ERP_BLUEPRINT §3.4).

needs departments, employees, users

used by application_guardians, attendance, candidates, class_sections, class_subject_teacher, departments and 11 more

COLUMN	TYPE	NOTES
user_id	bigint	→ users.id , unique
employee_code	varchar(16)	unique
employee_type	varchar(14)	
qualification	varchar(120)	optional
joining_date	date	optional
department_id	bigint	→ departments.id , optional
designation	varchar(60)	optional
reporting_to_id	bigint	→ employees.id , optional
status	varchar(13)	
exited_on	date	optional
emergency_contact_name	varchar(120)	optional
emergency_contact_phone	varchar(20)	optional
pan	varchar(10)	optional
uan	varchar(12)	optional

departments 6 rows

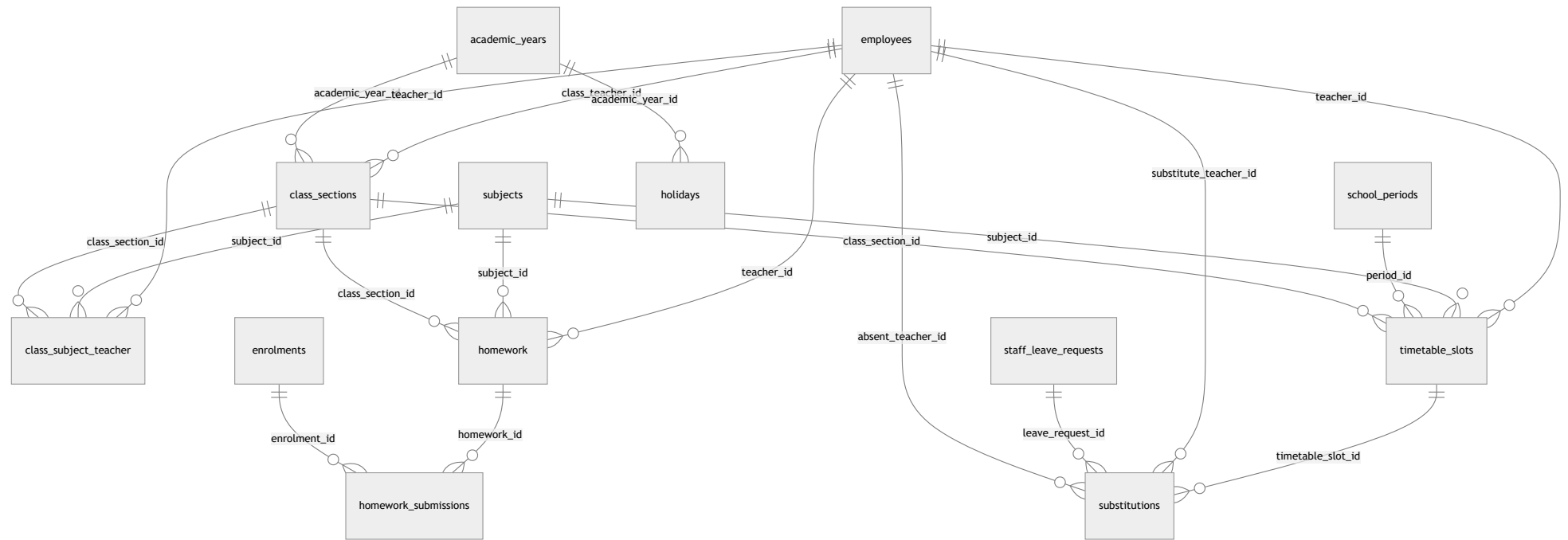
A teaching or administrative department (§5.3.3).

needs employees **used by** candidate_offers, employees

COLUMN	TYPE	NOTES
code	varchar(12)	unique
name	varchar(80)	
head_employee_id	bigint	→ employees.id , optional

COLUMN	TYPE	NOTES
esi_number	varchar(20)	optional
bank_account_no	varchar(20)	optional
bank_ifsc	varchar(11)	optional
bank_name	varchar(80)	optional

Academics



class_sections 10 rows

One class-and-section for one academic year - 10-A, 9-B - with its capacity, room and class teacher. The unit almost everything academic is scoped to.

needs academic_years, employees

used by class_subject_teacher, enrolments, exam_schedule, homework, notices, timetable_slots

COLUMN	TYPE	NOTES
class_name	varchar(8)	unique
section	varchar(4)	unique
class_teacher_id	bigint	→ employees.id, optional
academic_year_id	bigint	→ academic_years.id, unique
capacity	integer	optional
stream	varchar(20)	optional
room	varchar(20)	optional

subjects 9 rows

The catalogue of subjects the school teaches. A subject exists once and is attached to sections through `class\_subject\_teacher`.

used by class_subject_teacher, exam_schedule, homework, timetable_slots

COLUMN	TYPE	NOTES
name	varchar(60)	
code	varchar(12)	unique

class_subject_teacher 60 rows

Which teacher teaches which subject to which section. The join that makes a timetable solvable and decides whose marks entry is allowed.

needs class_sections, employees, subjects

COLUMN	TYPE	NOTES
class_section_id	bigint	→ class_sections.id , unique
subject_id	bigint	→ subjects.id , unique
teacher_id	bigint	→ employees.id

school_periods 6 rows

Bell timings, once per school instead of once per slot.

used by timetable_slots

COLUMN	TYPE	NOTES
period_no	integer	unique
start_time	time without time zone	
end_time	time without time zone	
name	varchar(20)	optional
is_break	boolean	

timetable_slots 300 rows

The weekly grid. Which subject, taught by which teacher, occupies which period on which weekday for which section.

needs class_sections, employees, school_periods, subjects

used by substitutions

COLUMN	TYPE	NOTES
class_section_id	bigint	→ class_sections.id , unique
day_of_week	varchar(3)	unique
period_id	bigint	→ school_periods.id , unique
subject_id	bigint	→ subjects.id
teacher_id	bigint	→ employees.id
room	varchar(20)	optional

substitutions 1 rows

One period, one day, covered by somebody else (§5.7.9).

needs employees, staff_leave_requests, timetable_slots

COLUMN	TYPE	NOTES
timetable_slot_id	bigint	→ timetable_slots.id , unique
date	date	unique
absent_teacher_id	bigint	→ employees.id
substitute_teacher_id	bigint	→ employees.id , optional
reason	text	
status	varchar(11)	
leave_request_id	bigint	→ staff_leave_requests.id , optional

holidays 5 rows

A day the school is shut.

needs academic_years

COLUMN	TYPE	NOTES
academic_year_id	bigint	→ academic_years.id , unique
date	date	unique
name	varchar(80)	

homework 8 rows

A piece of homework set for a section and subject, with the date it is due.

needs class_sections, employees, subjects **used by** homework_submissions

COLUMN	TYPE	NOTES
class_section_id	bigint	→ class_sections.id
subject_id	bigint	→ subjects.id
teacher_id	bigint	→ employees.id
title	varchar(160)	
description	text	optional
assigned_date	date	
due_date	date	
attachment_url	varchar(500)	optional

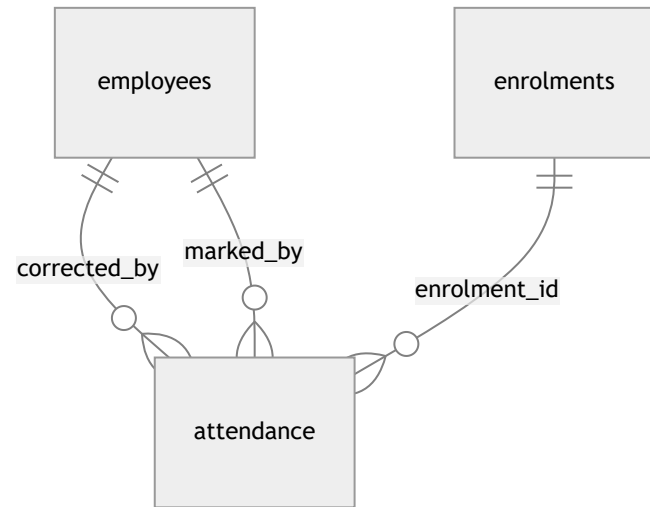
homework_submissions 47 rows

A child's submission against one `homework` row, and whether it has been marked.

needs enrolments, homework

COLUMN	TYPE	NOTES
homework_id	bigint	→ <code>homework.id</code> , unique
enrolment_id	bigint	→ <code>enrolments.id</code> , unique
answer_text	text	
submitted_at	timestampz	
marks	numeric(5,2)	optional
remarks	varchar(200)	optional
graded_at	timestampz	optional
attachment_url	varchar(500)	optional

Attendance



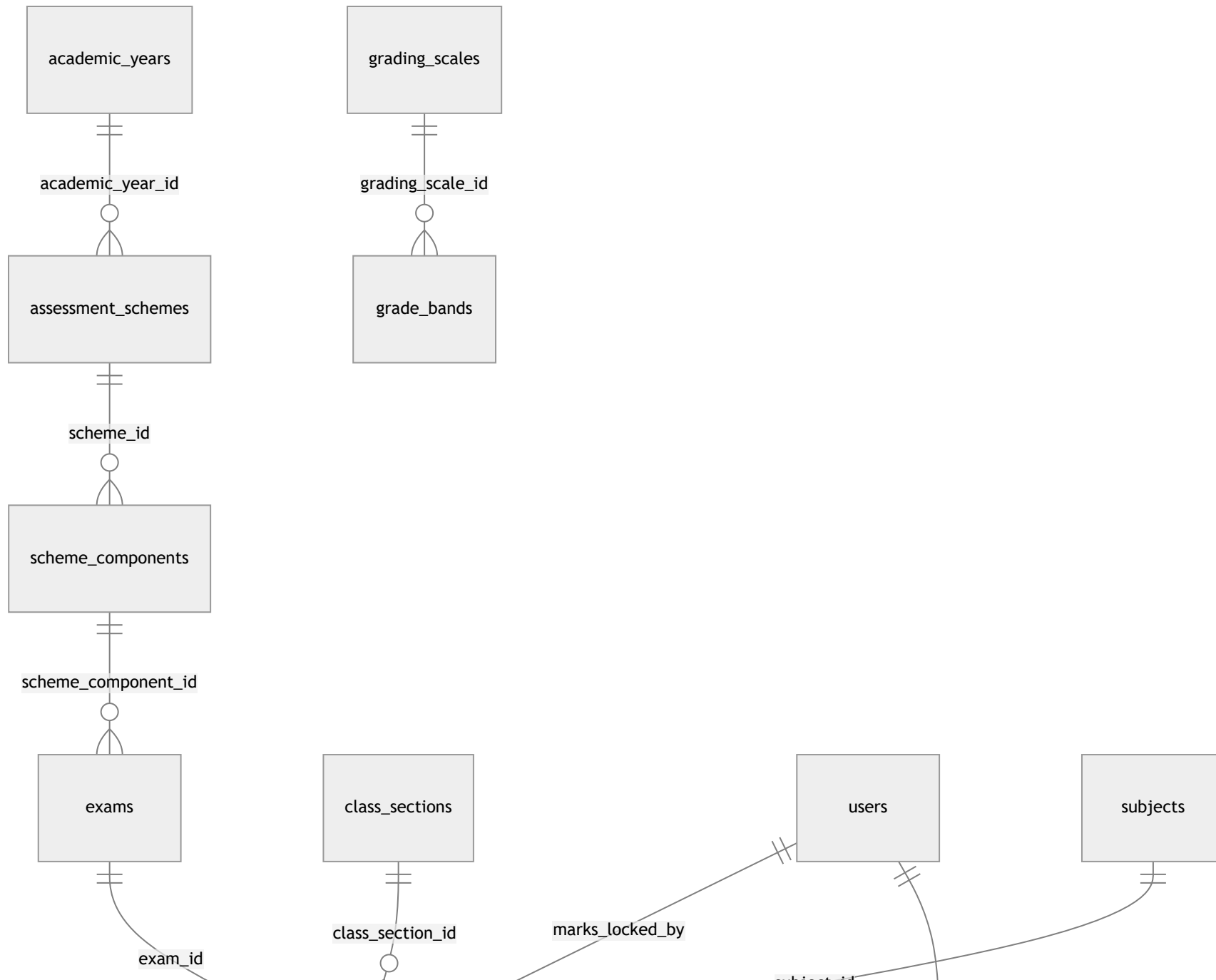
attendance 5,800 rows

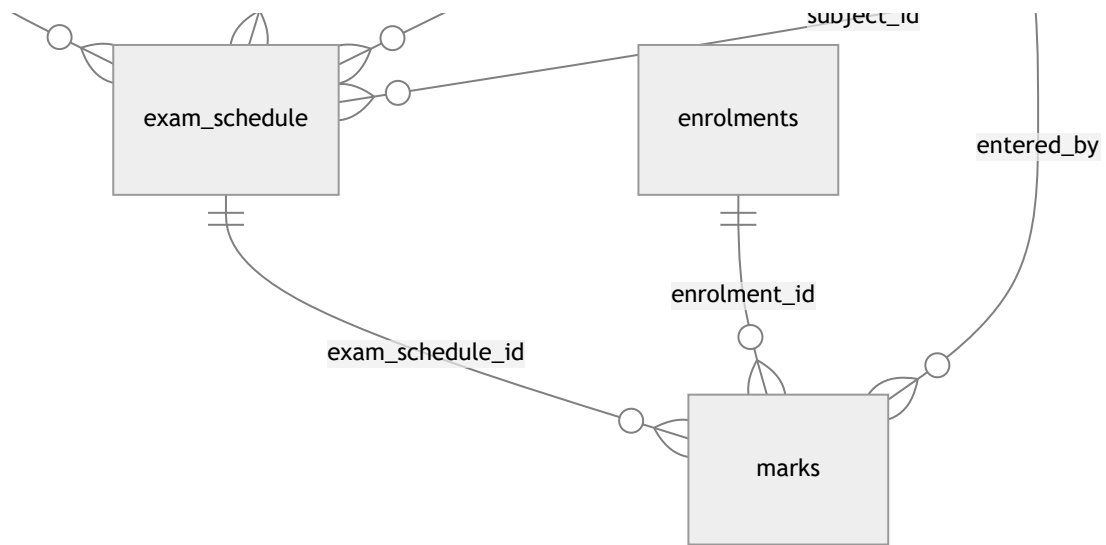
One day, one child, one mark.

needs employees, enrolments

COLUMN	TYPE	NOTES
enrolment_id	bigint	→ enrolments.id, unique
date	date	unique
status	varchar(8)	
marked_by	bigint	→ employees.id, optional
remarks	varchar(200)	optional
corrected_by	bigint	→ employees.id, optional
corrected_at	timestampz	optional

Examinations and results





exams 4 rows

One assessment event: "Term 1 Periodic Test", "Term 1 Examination".

needs scheme_components **used by** exam_schedule

COLUMN	TYPE	NOTES
name	varchar(120)	
term	varchar(20)	
start_date	date	
end_date	date	
scheme_component_id	bigint	→ scheme_components.id , optional

exam_schedule 240 rows

One paper: this exam, this section, this subject.

needs class_sections, exams, subjects, users **used by** marks

COLUMN	TYPE	NOTES
exam_id	bigint	→ exams.id , unique
class_section_id	bigint	→ class_sections.id , unique
subject_id	bigint	→ subjects.id , unique
exam_date	date	
start_time	time without time zone	optional
max_marks	numeric(5,2)	
room	varchar(20)	optional
marks_locked_at	timestampz	optional
marks_locked_by	bigint	→ users.id , optional

marks 2,400 rows

One child's result on one paper.

needs enrolments, exam_schedule, users

COLUMN	TYPE	NOTES
exam_schedule_id	bigint	→ exam_schedule.id, unique
enrolment_id	bigint	→ enrolments.id, unique
marks_obtained	numeric(5,2)	optional
is_absent	boolean	
is_exempted	boolean	
remarks	varchar(200)	optional
entered_by	bigint	→ users.id

assessment_schemes 1 rows

The shape of a year's assessment: its terms and what each is marked out of (§5.4.3).

needs academic_years

used by report_card_publications, scheme_components

COLUMN	TYPE	NOTES
academic_year_id	bigint	→ academic_years.id, unique
name	varchar(80)	unique
is_active	boolean	

scheme_components 8 rows

One markable column of the report card: "Term 1, Periodic Test, out of 10".

needs assessment_schemes used by exams

COLUMN	TYPE	NOTES
scheme_id	bigint	→ assessment_schemes.id , unique
term	varchar(20)	unique
code	varchar(12)	unique
name	varchar(60)	
max_marks	numeric(5,2)	
sequence	integer	

grading_scales 1 rows

A named set of grade bands, versioned.

used by grade_bands, report_card_publications

COLUMN	TYPE	NOTES
name	varchar(60)	unique
version	integer	unique
is_active	boolean	
frozen_at	timestampz	optional

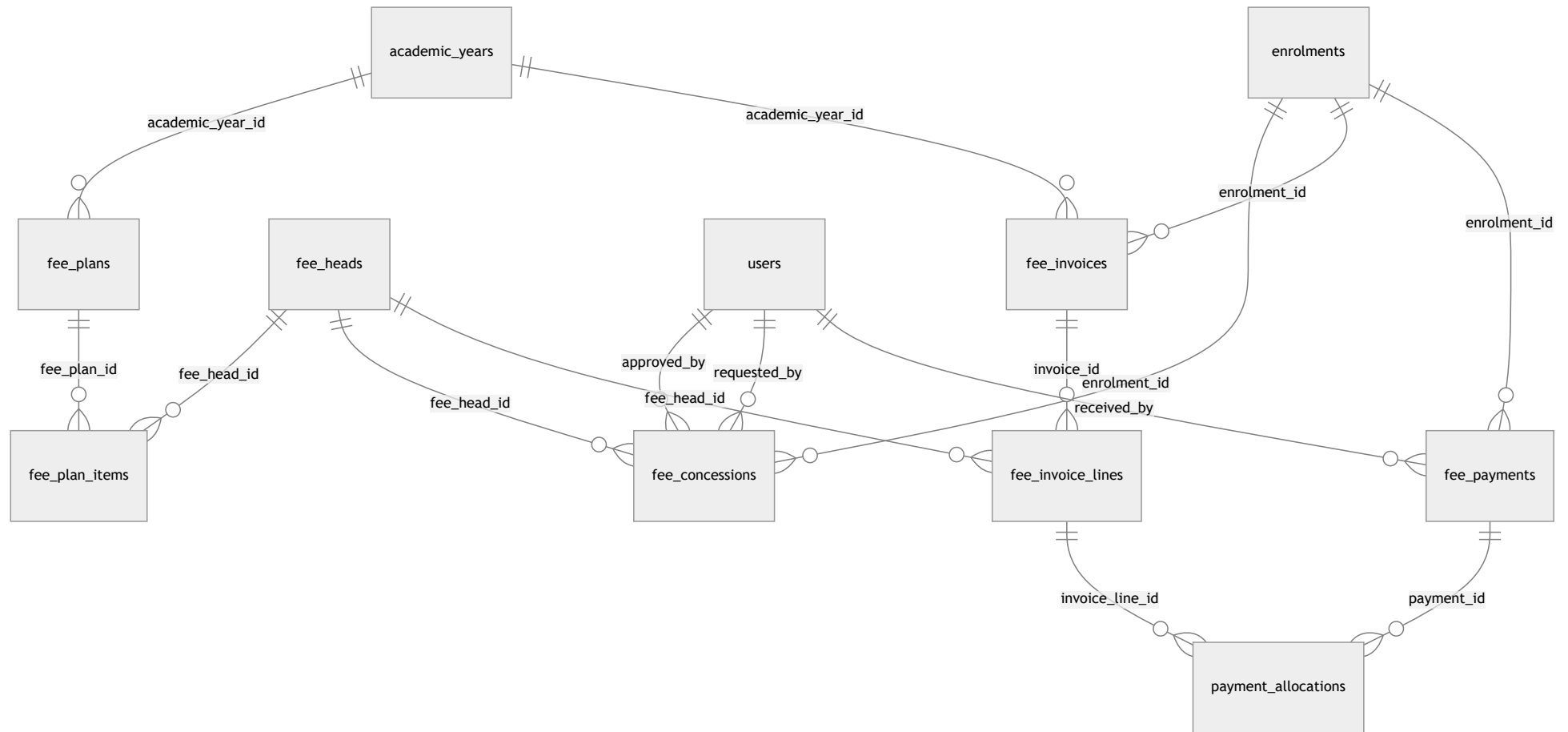
grade_bands 8 rows

One row of a grading scale: "91 and above is an A1".

needs grading_scales

COLUMN	TYPE	NOTES
grading_scale_id	bigint	→ grading_scales.id , unique
min_percent	numeric(5,2)	unique
grade	varchar(4)	unique
description	varchar(40)	optional

Fees and money



fee_heads 5 rows

The named charges a school bills: tuition, transport, exam fee. The vocabulary every invoice line is built from.

used by fee_concessions, fee_invoice_lines, fee_plan_items

COLUMN	TYPE	NOTES
name	varchar(60)	
code	varchar(16)	unique
type	varchar(9)	
is_refundable	boolean	
gl_code	varchar(20)	optional
is_active	boolean	

fee_plans 10 rows

What a class is charged for a year.

needs academic_years used by fee_plan_items, student_fee_plans

COLUMN	TYPE	NOTES
academic_year_id	bigint	→ academic_years.id, unique
name	varchar(60)	unique
class_name	varchar(8)	optional
is_active	boolean	

fee_plan_items 30 rows

The lines of a fee plan - which head, how much, how often. A plan means nothing without these.

needs fee_heads, fee_plans

COLUMN	TYPE	NOTES
fee_plan_id	bigint	→ fee_plans.id, unique
fee_head_id	bigint	→ fee_heads.id, unique
amount	numeric(10,2)	
frequency	varchar(8)	

fee_concessions 2 rows

A discount that someone approved.

needs enrolments, fee_heads, users

COLUMN	TYPE	NOTES
enrolment_id	bigint	→ enrolments.id
type	varchar(11)	
fee_head_id	bigint	→ fee_heads.id, optional
percent	numeric(5,2)	optional
amount	numeric(10,2)	optional
reason	text	
status	varchar(9)	
requested_by	bigint	→ users.id, optional
approved_by	bigint	→ users.id, optional
decided_at	timestamptz	optional
valid_from	date	optional
valid_to	date	optional

fee_invoices 300 rows

One month's bill for one enrolment (\$0.6: monthly, per student).

needs academic_years, enrolments **used by** fee_invoice_lines

COLUMN	TYPE	NOTES
enrolment_id	bigint	→ enrolments.id
academic_year_id	bigint	→ academic_years.id
invoice_no	varchar(24)	unique
period_month	integer	
period_year	integer	
issued_on	date	
due_date	date	
status	varchar(14)	
settled_on	date	optional
void_reason	text	optional

fee_invoice_lines 900 rows

One head on one invoice. The unit a payment allocates against.

needs fee_heads, fee_invoices **used by** payment_allocations

COLUMN	TYPE	NOTES
invoice_id	bigint	→ fee_invoices.id
fee_head_id	bigint	→ fee_heads.id
description	varchar(80)	
amount	numeric(10,2)	
discount	numeric(10,2)	

fee_payments 170 rows

Money received. Against an enrolment, not against an invoice.

needs enrolments, fee_payments, users **used by** payment_allocations

COLUMN	TYPE	NOTES
enrolment_id	bigint	→ enrolments.id
receipt_no	varchar(24)	unique
amount	numeric(10,2)	
method	varchar(20)	
instrument_ref	varchar(40)	optional
received_at	timestampz	
received_by	bigint	→ users.id, optional
idempotency_key	varchar(64)	unique
status	varchar(8)	
reverses_payment_id	bigint	→ fee_payments.id, optional
reason	text	optional

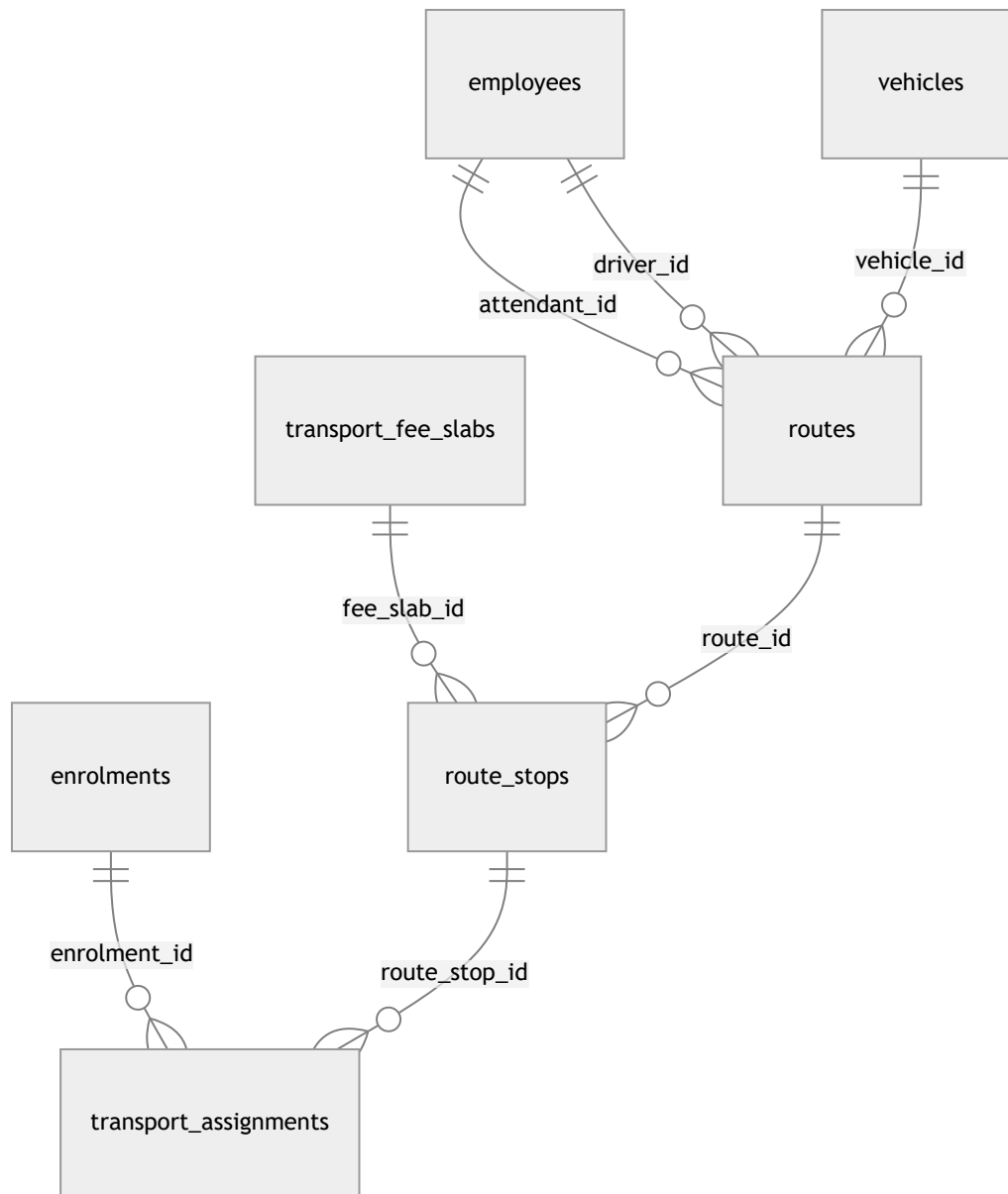
payment_allocations 450 rows

How much of a payment settled which line. The ledger's only truth about what is paid — every balance in the system is a SUM over this table.

needs fee_invoice_lines, fee_payments

COLUMN	TYPE	NOTES
payment_id	bigint	→ fee_payments.id
invoice_line_id	bigint	→ fee_invoice_lines.id
amount	numeric(10,2)	

Transport



vehicles 2 rows

A bus. `capacity` is the seating figure on the registration papers, and it is the number the hard block in §5.6.9 counts against — not a soft target the office can talk its way past.

used by routes

COLUMN	TYPE	NOTES
registration_no	varchar(20)	unique
make_model	varchar(60)	optional
capacity	integer	
ownership	varchar(5)	
status	varchar(17)	
gps_device_id	varchar(40)	optional

routes 2 rows

One bus doing one circuit.

needs employees, vehicles used by route_stops

COLUMN	TYPE	NOTES
code	varchar(16)	unique
name	varchar(80)	
vehicle_id	bigint	→ vehicles.id, optional
driver_id	bigint	→ employees.id, optional
attendant_id	bigint	→ employees.id, optional
distance_km	numeric(6,2)	optional
status	varchar(9)	

route_stops 7 rows

A halt on a route, with the times the bus is actually there.

needs routes, transport_fee_slabs **used by** transport_assignments

COLUMN	TYPE	NOTES
route_id	bigint	→ routes.id, unique
sequence	integer	unique
name	varchar(80)	
landmark	varchar(120)	optional
pickup_time	time without time zone	
drop_time	time without time zone	optional
fee_slab_id	bigint	→ transport_fee_slabs.id, optional
latitude	float	optional
longitude	float	optional

transport_assignments 30 rows

A child on a bus, from a date until a date.

needs enrolments, route_stops

COLUMN	TYPE	NOTES
enrolment_id	bigint	→ enrolments.id
route_stop_id	bigint	→ route_stops.id
direction	varchar(6)	
start_date	date	
end_date	date	optional
status	varchar(9)	
end_reason	text	optional

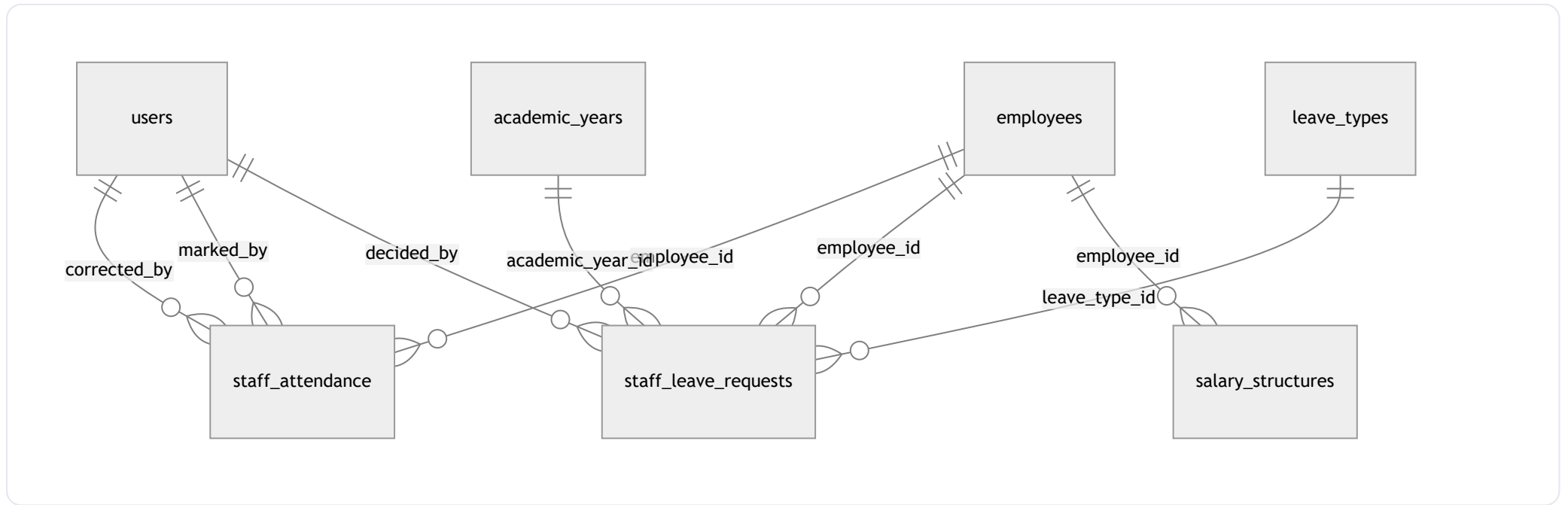
transport_fee_slabs 3 rows

A distance band and its monthly price.

used by route_stops

COLUMN	TYPE	NOTES
name	varchar(40)	unique
monthly_amount	numeric(10,2)	
is_active	boolean	

HR and payroll



staff_attendance 1 rows

One day, one member of staff, one mark.

needs employees, users

COLUMN	TYPE	NOTES
employee_id	bigint	→ employees.id, unique
date	date	unique
status	varchar(8)	
check_in	time without time zone	optional
check_out	time without time zone	optional
marked_by	bigint	→ users.id, optional
corrected_by	bigint	→ users.id, optional
remarks	varchar(200)	optional

staff_leave_requests 1 rows

One application, from draft to decision.

needs academic_years, employees, leave_types, users used by substitutions

COLUMN	TYPE	NOTES
employee_id	bigint	→ employees.id
leave_type_id	bigint	→ leave_types.id, optional
academic_year_id	bigint	→ academic_years.id
from_date	date	
to_date	date	
is_half_day	boolean	
days	numeric(5,1)	
reason	text	
status	varchar(9)	
balance_exception	boolean	
decided_by	bigint	→ users.id, optional
decided_at	timestamptz	optional
decision_note	text	optional

leave_types 4 rows

One kind of staff leave, with its yearly entitlement.

used by leave_balances, staff_leave_requests

COLUMN	TYPE	NOTES
code	varchar(12)	unique
name	varchar(60)	
annual_quota	numeric(5,1)	
is_paid	boolean	
active	boolean	

salary_components 11 rows

One line a payslip can carry, and how its amount is worked out.

used by payslip_lines, salary_structure_items

COLUMN	TYPE	NOTES
code	varchar(16)	unique
name	varchar(60)	
type	varchar(21)	
calculation	varchar(16)	
value	numeric(12,2)	
applies_below_gross	numeric(12,2)	optional
taxable	boolean	
statutory	boolean	
active	boolean	
sequence	integer	

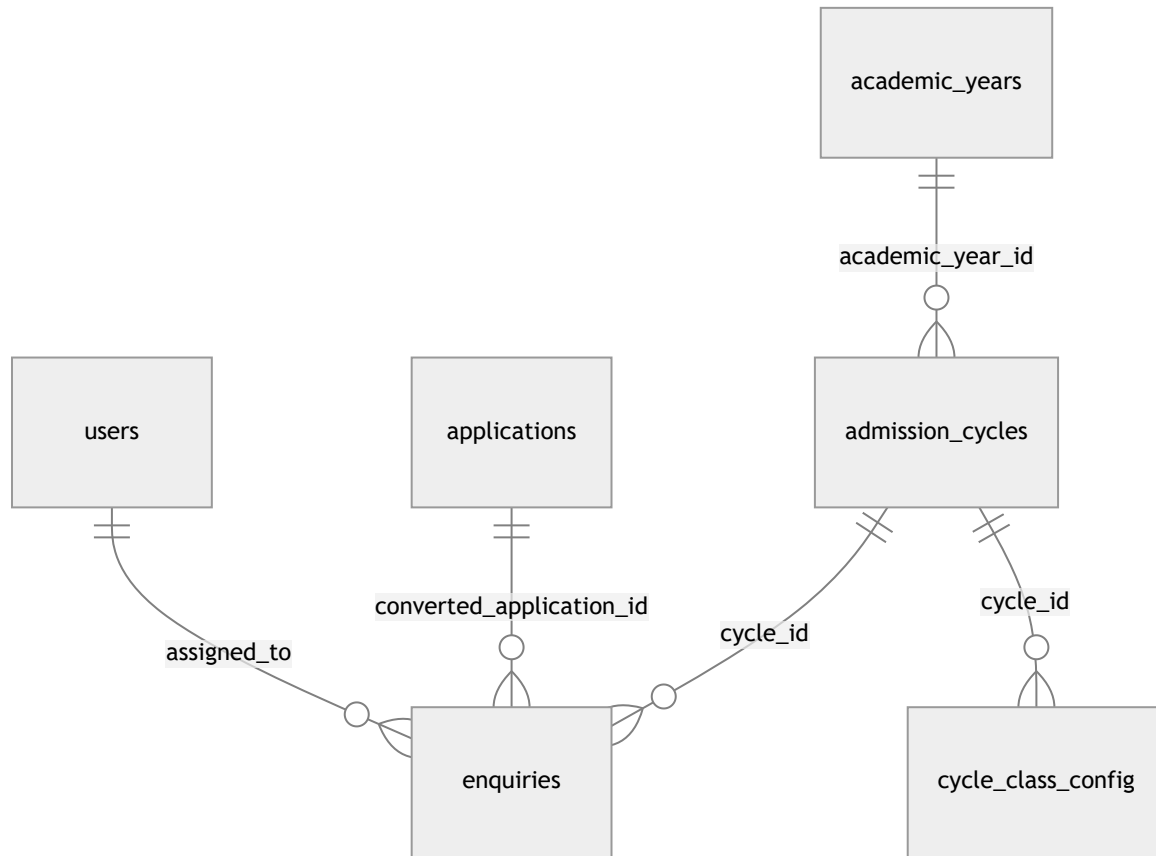
salary_structures 16 rows

What one employee is on, from a date.

needs employees used by salary_structure_items

COLUMN	TYPE	NOTES
employee_id	bigint	→ employees.id, unique
effective_from	date	unique
monthly_gross	numeric(12,2)	
is_active	boolean	
note	text	optional

Admission



enquiries 6 rows

A prospective parent's enquiry, before any application exists. The top of the admission funnel.

needs admission_cycles, applications, users **used by** enquiry_interactions

COLUMN	TYPE	NOTES
cycle_id	bigint	→ admission_cycles.id
enquirer_name	varchar(120)	
mobile	varchar(20)	
email	varchar(160)	optional
child_name	varchar(120)	optional
child_dob	date	optional
class_of_interest	varchar(8)	optional
source	varchar(10)	
status	varchar(23)	
assigned_to	bigint	→ users.id, optional
next_follow_up_on	date	optional
converted_application_id	bigint	→ applications.id, optional

admission_cycles 1 rows

One intake season for one academic year.

needs academic_years

used by applications, cycle_class_config, enquiries, waitlist_entries

COLUMN	TYPE	NOTES
academic_year_id	bigint	→ academic_years.id, unique
name	varchar(80)	unique
status	varchar(8)	
starts_on	date	optional
ends_on	date	optional
application_fee	numeric(10,2)	
late_fee	numeric(10,2)	
allow_online_applications	boolean	
admission_fee_refund_policy	text	optional

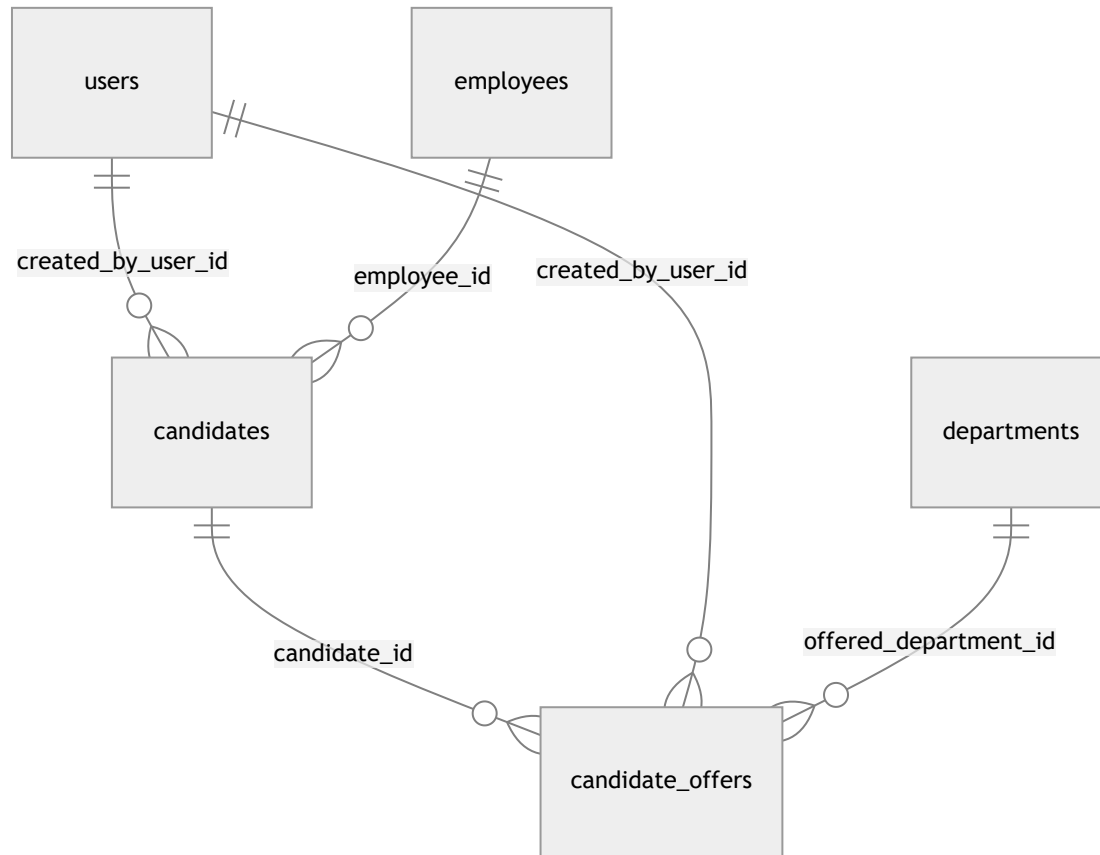
cycle_class_config 3 rows

Seats and rules for one class within one cycle.

needs admission_cycles

COLUMN	TYPE	NOTES
cycle_id	bigint	→ admission_cycles.id, unique
class_name	varchar(8)	unique
stream	varchar(20)	unique, optional
total_seats	integer	
reserved_seats	json	optional
age_on	date	optional
min_age_years	numeric(4,2)	optional
max_age_years	numeric(4,2)	optional
requires_test	boolean	
requires_interview	boolean	
required_document_codes	json	optional

Teacher recruitment and hiring



candidates 1 rows

A teacher applicant registered for recruitment.

needs employees, users **used by** candidate_offers

COLUMN	TYPE	NOTES
application_no	varchar(50)	unique
full_name	varchar(120)	
email	varchar(255)	
phone	varchar(20)	
gender	varchar(20)	
date_of_birth	date	
address	text	optional
post_applied_for	varchar(100)	
qualification	varchar(200)	
specialization	varchar(100)	optional
experience_years	numeric(4,1)	
previous_school	varchar(200)	optional
expected_salary	numeric(12,2)	optional
resume_file_key	varchar(500)	optional
photo_file_key	varchar(500)	optional
status	varchar(11)	

candidate_offers 1 rows

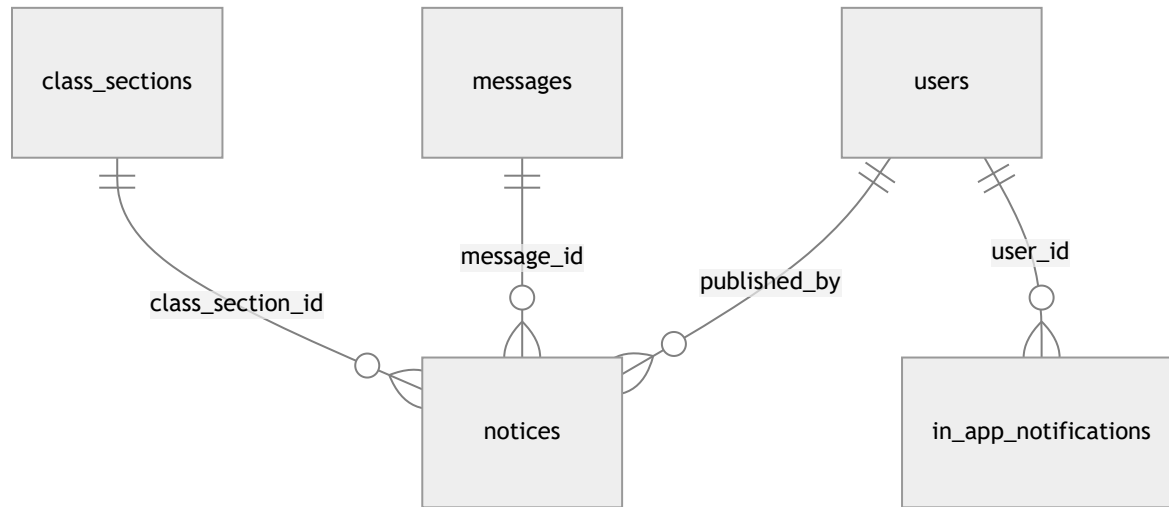
Job offer terms extended to a selected candidate.

needs candidates, departments, users

COLUMN	TYPE	NOTES
candidate_id	bigint	→ candidates.id , unique
offered_designation	varchar(100)	
offered_department_id	bigint	→ departments.id , optional
offered_salary	numeric(12,2)	
joining_date	date	
offer_notes	text	optional
created_by_user_id	bigint	→ users.id , optional

COLUMN	TYPE	NOTES
notes	text	optional
employee_id	bigint	→ employees.id , unique, optional
created_by_user_id	bigint	→ users.id , optional

Communication



notices 6 rows

The noticeboard. An announcement with an audience - the whole school, one class, parents, teachers - optionally also sent as a message.

needs class_sections, messages, users

COLUMN	TYPE	NOTES
title	varchar(160)	
body	text	
audience	varchar(8)	
class_section_id	bigint	→ class_sections.id , optional
published_by	bigint	→ users.id
published_at	timestampz	
message_id	bigint	→ messages.id , optional

message_templates 4 rows

Versioned, so the exact text sent stays reproducible (§5.9.9).

used by messages

COLUMN	TYPE	NOTES
code	varchar(40)	unique
version	integer	unique
name	varchar(80)	
channel	varchar(8)	
category	varchar(11)	
subject	varchar(200)	
body	text	
is_active	boolean	

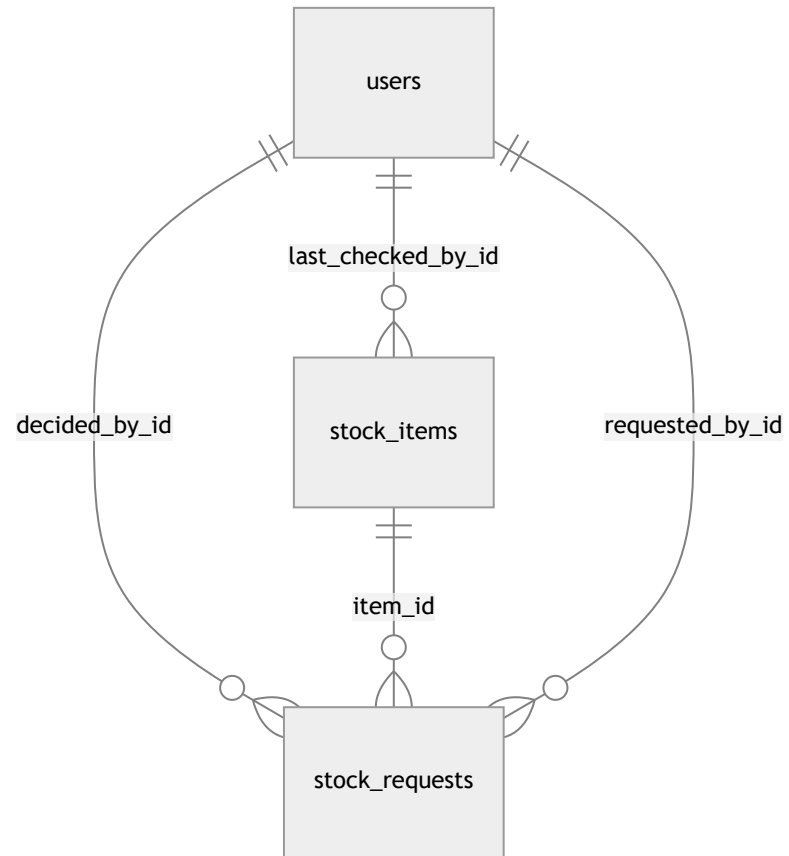
in_app_notifications 1 rows

In-app alert for staff, teachers, or admins regarding leaves, substitutions, etc.

needs users

COLUMN	TYPE	NOTES
user_id	bigint	→ users.id
title	varchar(150)	
message	text	
category	varchar(50)	
link_url	varchar(255)	optional
is_read	boolean	

Inventory and stock



stock_items 31 rows

An inventory item held in a school store, lab, or room.

needs [users](#) **used by** [stock_requests](#)

COLUMN	TYPE	NOTES
name	varchar(120)	
category	varchar(60)	
location	varchar(80)	
unit	varchar(30)	
current_quantity	integer	
min_quantity	integer	
unit_cost	numeric(10,2)	optional
is_critical	boolean	
has_discrepancy	boolean	
discrepancy_notes	text	optional
last_checked_at	timestampz	optional
last_checked_by_id	bigint	→ users.id , optional

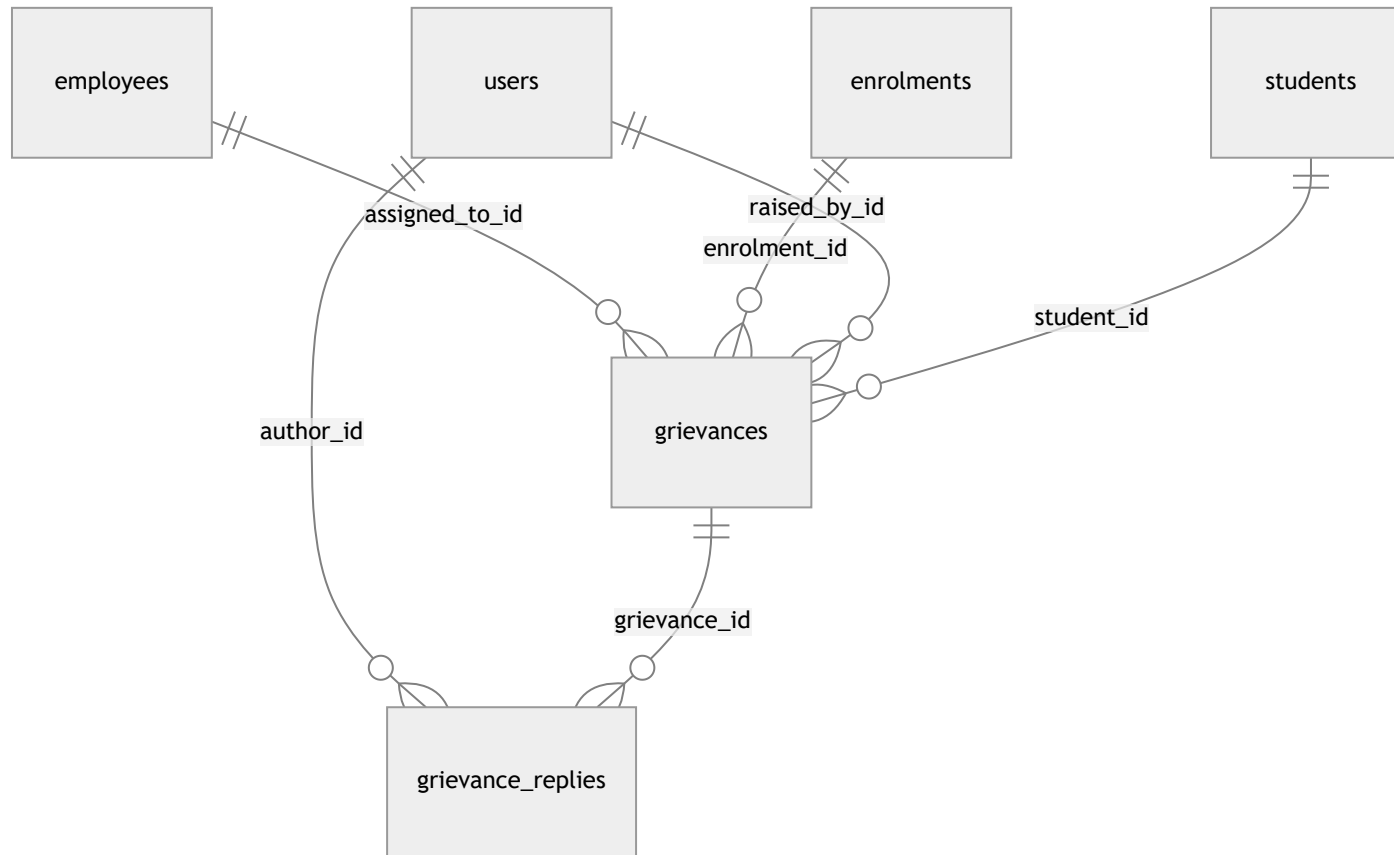
stock_requests 9 rows

A stock replenishment request, purchase indent, or teacher diminishing-stock flag.

needs [stock_items](#), [users](#)

COLUMN	TYPE	NOTES
item_id	bigint	→ stock_items.id , optional
item_name	varchar(120)	
category	varchar(60)	
location	varchar(80)	
quantity_requested	integer	
urgency	varchar(20)	
status	varchar(20)	
flag_type	varchar(20)	
requested_by_id	bigint	→ users.id
requested_by_name	varchar(120)	
reason	text	
decided_by_id	bigint	→ users.id , optional
decided_at	timestampz	optional
decision_note	text	optional

Grievances and feedback



grievances 5 rows

A formal grievance or issue raised by a teacher, parent, or student.

needs employees, enrolments, students, users **used by** grievance_replies

COLUMN	TYPE	NOTES
title	varchar(200)	
description	text	
category	varchar(60)	
raised_by_id	bigint	→ users.id
raised_by_role	varchar(30)	
raised_by_name	varchar(120)	
student_id	bigint	→ students.id , optional
student_name	varchar(120)	optional
enrolment_id	bigint	→ enrolments.id , optional
status	varchar(30)	
priority	varchar(20)	
assigned_to_id	bigint	→ employees.id , optional
assigned_to_name	varchar(120)	optional
resolution_notes	text	optional
resolved_at	timestampz	optional

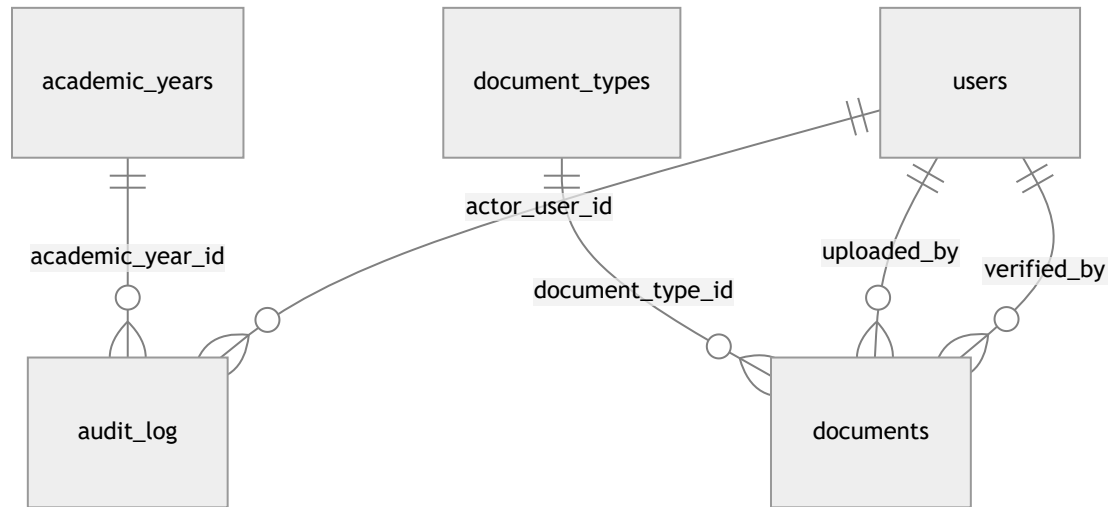
grievance_replies 4 rows

A response or progress note on a grievance thread.

needs grievances, users

COLUMN	TYPE	NOTES
grievance_id	bigint	→ grievances.id
author_id	bigint	→ users.id
author_name	varchar(120)	
author_role	varchar(30)	
message	text	
is_internal	boolean	

System



audit_log 263 rows

Append-only. No update path, no delete path, never truncated.

needs academic_years, users

COLUMN	TYPE	NOTES
occurred_at	timestampz	
actor_user_id	bigint	→ users.id, optional
actor_label	varchar(120)	optional
entity_type	varchar(40)	
entity_id	bigint	optional
action	varchar(13)	
before	json	optional
after	json	optional
reason	text	optional
academic_year_id	bigint	→ academic_years.id, optional
ip	varchar(45)	optional

documents 13 rows

An uploaded file attached to a student, an employee or a vehicle, with its verification state and expiry. Deleted softly, so the audit entry still points at something.

needs document_types, users

COLUMN	TYPE	NOTES
owner_type	varchar(11)	
owner_id	bigint	
document_type_id	bigint	→ document_types.id, optional
file_key	varchar(255)	unique
file_name	varchar(255)	
mime_type	varchar(120)	
size_bytes	bigint	
checksum	varchar(64)	optional
uploaded_by	bigint	→ users.id, optional
uploaded_at	timestampz	
status	varchar(17)	
original_seen	boolean	
verified_by	bigint	→ users.id, optional
verified_at	timestampz	optional
rejection_reason	text	optional

COLUMN	TYPE	NOTES
expires_on	date	optional
is_confidential	boolean	
deleted_at	timestamptz	optional
deleted_by	bigint	optional
delete_reason	text	optional

document_types 21 rows

Configurable, not an enum: required-document lists change by class, by category and by state regulation, and a school must be able to edit them without a deploy (§3.8).

used by documents

COLUMN	TYPE	NOTES
code	varchar(40)	unique
name	varchar(120)	
applies_to	varchar(11)	
is_mandatory	boolean	
required_if_category	varchar(40)	optional
has_expiry	boolean	
is_confidential	boolean	
sort_order	integer	

scheduled_jobs 4 rows

A recurring job. Deliberately not cron syntax — a school needs "every day at 02:00", not "* / 7 3-5 * * 2", and an interval plus an hour is far easier for a non-technical administrator to read in a settings screen.

COLUMN	TYPE	NOTES
kind	varchar(48)	unique
payload	json	optional
enabled	boolean	
every_minutes	integer	
at_hour	integer	optional
last_run_at	timestamptz	optional
next_run_at	timestamptz	optional

Diagrams read `parent ||--o{ child` , labelled with the column that carries the key. Edges leaving an area are drawn, because a key that crosses a boundary is the one that gets forgotten.

Columns, types, keys and row counts are read from the running database, not written by hand. Generated against migration `a1b2c3d4e5f6` on branch `slice/office-feedback` ; row counts are the seeded demo school's. Regenerate with `python scripts/gen_db_essentials.py` .